Carlos R. Salazar

PhD Student · IFM (Theoretical Physics)
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Education

Linköping University — *Linköping, Sweden*
PhD Theoretical Physics — *February 2024 - present*
- Supervisor: Prof. Davide Sangiovanni

Aix-Marseille Université — *Marseille, France*
MS, Fundamental Physics — *August 2021 - June 2023*
- Theis Supervisor: Dr. Julien Lam

Jacobs University Bremen — *Bremen, Germany*
BS, Physics — *September 2018 - June 2021*
- Minor in Computer Science
- Thesis Supervisor: Prof. Ulrich Kleinekathöfer

Professional Experience

2024 - Present	**Doctoral Candidate - Computational and Machine Learning Studies of Mechanical Properties of Alloys - 10% Teaching, 5% Doctoral Student Council**, Linköping University
2023	**Internship - Structural Analysis by Statistical Learning Method for Atomic Scale Nanoparticles**, UMET - ISP
2022	**Internship - Machine Learning Approaches to Model Interatomic Interactions in Materials**, CEMES - CNRS
2021	**Bachelor's Thesis - Molecular Dynamics Simulation of Porphyrin Adsorption on Clay for the formation of a Light Harvesting System**, Jacobs University Bremen
2021	**Teaching Assistant - Computational Physics Lab**, Jacobs University Bremen
2020	**Student Laboratory Assistant - Ocean Laboratory**, Jacobs University Bremen
2020	**Teaching Assistant - Classical Physics**, Jacobs University Bremen
2020	**Teaching Assistant - Physics for the Natural Sciences**, Jacobs University Bremen

Publications

Published

Salazar, C. R., Ammothum Kandy, A. K., Furstoss, J., Gromoff, Q., Goniakowski, J., & Lam, J. (2024). **Competing nucleation pathways in nanocrystal formation**. npj Comput. Mater., 10(1), 199.

Furstoss, J., Salazar, C. R., Carrez, P., Hirel, P., & Lam, J. (2025). **All-around local structure classification with supervised learning: The example of crystal phases and dislocations in complex oxides**. Comput. Phys. Commun., 309, 109480.

Lyu, D., Vinod, V., Holzenkamp, M., Holtkamp, Y. M., Maity, S., Salazar, C. R., ...Zaspel, P. (2025). **Excitation Energy Transfer between Porphyrin Dyes on a Clay Surface: A Study Employing Multifidelity Machine Learning**. Adv. Theor. Simul., 8(11), e00271.

Gromoff, Q., Benoit, M., Goniakowski, J., Salazar, C. R., & Lam, J. (2026). **Growth driven phase transitions in Zinc Oxide nanoparticles through machine-learning assisted simulations**. Nanoscale, 18(24), 12903–12912.

Preprints

Salazar, C. R., Ghosh, S., Tasnádi, F., Johnson, L. J. S., Lam, J., Furstoss, J., & Sangiovanni, D. G. (2026). **Bonding adaptability and transformation plasticity in Ti–Al–Si–N alloys**. SSRN Preprint 7038127.

Tasnádi, F., Wästlund, J., Ghosh, S., Gómez Jaramillo, S., Salazar, C. R., Sangiovanni, D. G., ...Abrikosov, I. A. (2026). **HADB Database: From Data Generation to AI-Supported Predictions of Properties of Hard-Coating Alloy**. SSRN Preprint 7228547.

Sangiovanni, D. G., Salazar, C. R., Tasnádi, F., Klarbring, J., & Abrikosov, I. A. (2026). **Concerted mass transport governs anomalous diffusion in anharmonic bcc metals**. Preprint.